UNIVERSITI TEKNOLOGI MARA
TEST

COURSE	:	**DIGITAL ELECTRONICS**
COURSE CODE	:	**ITT270**
NAME OF COURSE LECTURER	:	
UiTM STUDENT NAME	:	
UiTM STUDENT CARD NO	:	
FACULTY	:	**KPPIM**
PROGRAMME/CODE	:	**CS110/CS143**
PART	:	
GROUP	:	
TIME	:	**2 HOURS**

INSTRUCTIONS TO CANDIDATES

1. Fill in the particulars above.

2. Do not bring any material into the examination room unless permission is given by the invigilator.

3. Answer **ALL** questions

PART	QUESTION	MARKS
A	1 - 10	/20
B	1	/4
	2	/6
	3	/7
	4	/3
	5	/6
	6	/4
	TOTAL	/50

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This test paper consists of 5 printed pages

PART A (20 MARKS)

1. Why is Boolean algebra important in digital electronics?
   A. It converts analog signals into digital format.
   B. It helps design and simplify digital logic circuits.
   C. It stores large amounts of data.
   D. It measures the voltage of digital components.

2. In a digital voting system, a light should turn ON only when both voters agree (either both vote YES or both vote NO). Which logic gate is best suited for this condition?
   A. XOR gate
   B. AND gate
   C. OR gate
   D. XNOR gate

3. If a system is designed using the binary number system, what would the output of the logic gate "1101 XOR 1001" be?
   A. 0100
   B. 1111
   C. 1011
   D. 1001

4. Simplify the following Boolean expression using Boolean algebra laws:

   $Y = A \cdot B + A \cdot B'$

   A. A
   B. B
   C. A + B
   D. AB′

5. Which of the following Boolean expressions is in Sum of Products (SOP) form?
   A. (A + B)(C + D)
   B. A·B + C·D
   C. (A + B) + (C + D)
   D. A + B + C + D

6. Which of the following statements best describes the behavior of a NAND gate?

   A. The output is high only when both inputs are low.
   B. The output is high when both inputs are high.
   C. The output is high unless both inputs are high.
   D. The output is low when both inputs are low.

7. How many alphabetic characters are used in the hexadecimal number system?
   A. 6
   B. 7
   C. 15
   D. 16

8. Which of the following Boolean expressions is the complement of the expression:

   Y = (A + B) · (A' + C)

   A. A' + B' + A · C
   B. A · B' + A' · C
   C. A' . B’ + A . C'
   D. A · (B + C')

9. What is the hexadecimal representation of the BCD number 000110010111_{BCD}?
   A. 197
   B. C5
   C. 11000101
   D. 407

10. Using De Morgan's Theorems, how can you express $\overline{\boldsymbol{A}.\boldsymbol{B}} + \overline{\boldsymbol{C}}.\overline{\boldsymbol{D}}$?
    A. $\overline{\boldsymbol{A}.\boldsymbol{B}.\boldsymbol{C}.\boldsymbol{D}}$
    B. $\overline{\boldsymbol{A}}.\overline{\boldsymbol{B}} + \overline{\boldsymbol{C}}.\overline{\boldsymbol{D}}$
    C. $\overline{\boldsymbol{A}} + \overline{\boldsymbol{B}} + \overline{\boldsymbol{C}}.\overline{\boldsymbol{D}}$
    D. $\overline{\boldsymbol{A}} + \overline{\boldsymbol{B}} + \overline{\boldsymbol{C}} + \overline{\boldsymbol{D}}$

PART B (30 MARKS)

QUESTION 1

a) Define digital electronics and state one of its applications.

(2 marks)

b) State two common challenges faced in the field of digital electronics.

(2 marks)

QUESTION 2

Convert the following numbers (show your work):

a) 10010101_{BCD} to hexadecimal

(2 marks)

b) 41576_8 to binary

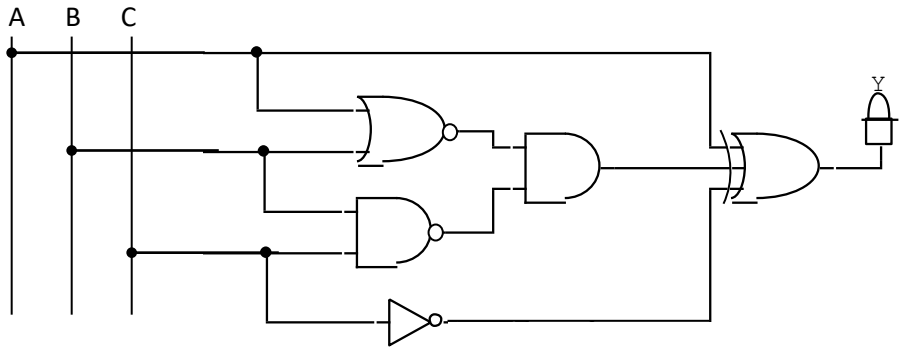
(2 marks)

c) 101110110_2 into decimal.

(2 marks)

QUESTION 3

Given the following logic circuit:



a) Write a Boolean expression.

(2 marks)

b) Derive the truth table.

(3 marks)

c) Write the product-of-sum (POS) expressions for Y

(2 marks)

QUESTION 4

Simplify the following equation using Boolean Theorems and DeMorgan Theorem:

$$Y = \overline{\mathrm{A}.\mathrm{B}} + \overline{\mathrm{B}}.\mathrm{C} + (\overline{\mathrm{A}} + \mathrm{B}).\overline{\mathrm{C}}$$

(3 marks)

QUESTION 5

Given the following Boolean expression

$$\mathrm{Y} = \left(\overline{\mathrm{A}}.\mathrm{B}\right)(\overline{\mathrm{B} + \mathrm{C}}) + \overline{\mathrm{B}}.\mathrm{C}$$

Draw the logic circuit using

a) NAND gates only

(3 marks)

b) NOR gates only

(3 marks)

QUESTION 6

In a smart home, a digital logic system is used to control two features in a room based on three inputs:

A: Motion detected (1 = motion, 0 = no motion)
B: Daylight present (1 = bright daylight, 0 = dark)
C: Manual switch status (1 = switch ON, 0 = switch OFF)

The system controls two outputs:

Light: The light should turn ON only when:

Motion is detected, it is dark, and the manual switch is OFF, OR, when the manual switch is ON, regardless of A and B.

Fan: The fan should turn ON only when:

Motion is detected AND either the manual switch is ON or it is daylight.

Derive the truth table for all possible combinations of inputs and their corresponding outputs

(4 marks)

END OF QUESTION